Day 3 - NI TestStand sequences

Reuse the Day 2 SubVIs as TestStand code modules. Build the LM321 DC test one sequence at a time, run it after each addition, and keep cleanup in the Cleanup group.

1. Before you start

1.1 Open the Day 2 LabVIEW project. Confirm that **pps_connect**, **pps_ctrl**, **dmm_connect**, **dmm_read**, **switch_init**, **switch_ctrl**, **ao_init**, **ao_update**, **ai_init**, **ai_measure**, **daqmx_stop** and **visa_close** run on this station.

1.2 Open NI TestStand 2026 Q1. Under **Configure > Adapters**, select the LabVIEW adapter and the LabVIEW version used for the Day 2 VIs.

1.3 Create a folder for today's work. Save every sequence file (.seq) there and keep the VI paths relative to it.

1.4 Keep the DUT and AUX supplies off until a step asks for them.

2. Working rules

2.1 **Setup** opens resources, **Main** applies stimuli and measures, **Cleanup** returns the station to a safe state and releases resources. Cleanup also runs after a failure or Terminate.

2.2 Store instrument references in **FileGlobals** so all sequences in the file can use them. Store measurements and operator inputs in **Locals**.

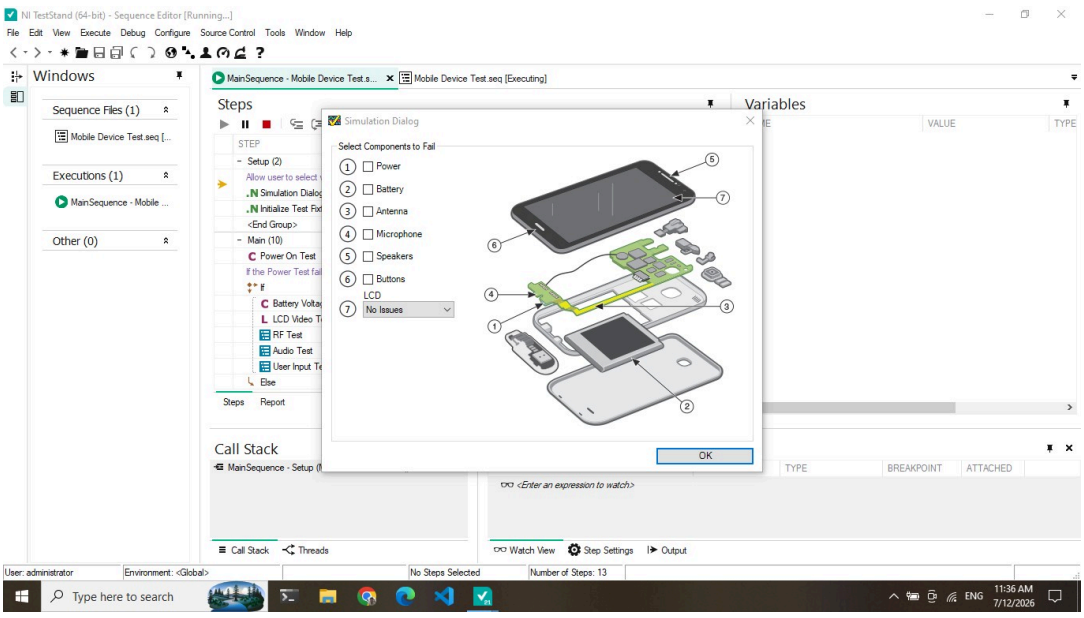
2.3 Use an **Action** step when no limit is evaluated. Use a **Numeric Limit Test** when a measurement decides PASS/FAIL.

2.4 Run with a breakpoint and **Step Over** after each change. Check variable values in the **Variables** pane.

3. Try a ready-made example

3.1 Open **Examples > Demos > Mobile Device Test** and choose **Execute > Run MainSequence**.

3.2 Set a breakpoint on one step, run again, and use **Step Into** and **Step Over**. Note how each step reports its status.



Tutorial 1: Accept and validate the supply voltage

Goal: Read a value from the operator, store it in a variable and branch on its range.

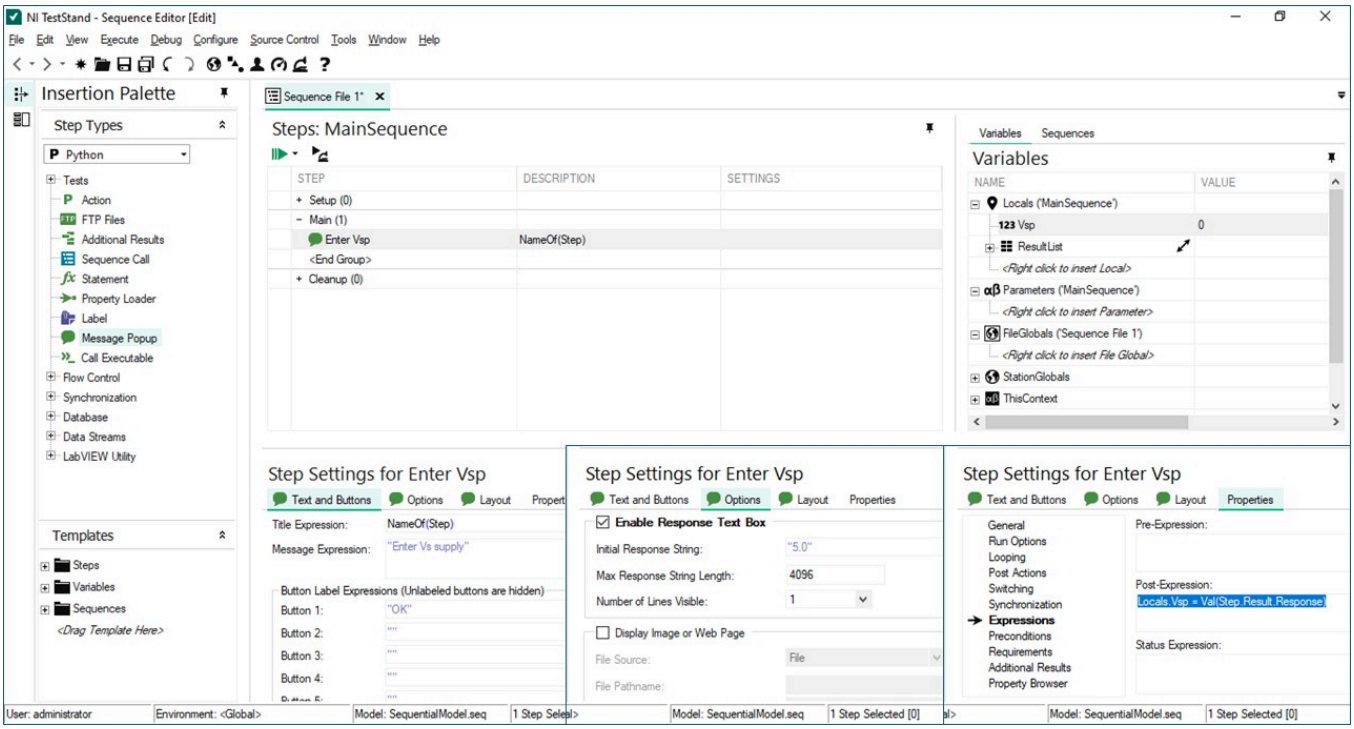


Figure 1. A Message Popup with a response text box; the Post-Expression converts the response to a number.

1. Read the value

1.1 Create a new sequence file and save it as **Day3_Tutorial1.seq**.

1.2 In **MainSequence > Main**, add a **Message Popup** with the message “Enter Vs supply”. Enable **Enable Response Text Box**.

1.3 Add the Number variable **Locals.Vsp**.

1.4 In **Properties > Expressions > Post-Expression**, enter `Locals.Vsp = Val(Step.Result.Response)`.

1.5 Set a breakpoint after the popup. Run, enter 15 and check that `Locals.Vsp` is 15.

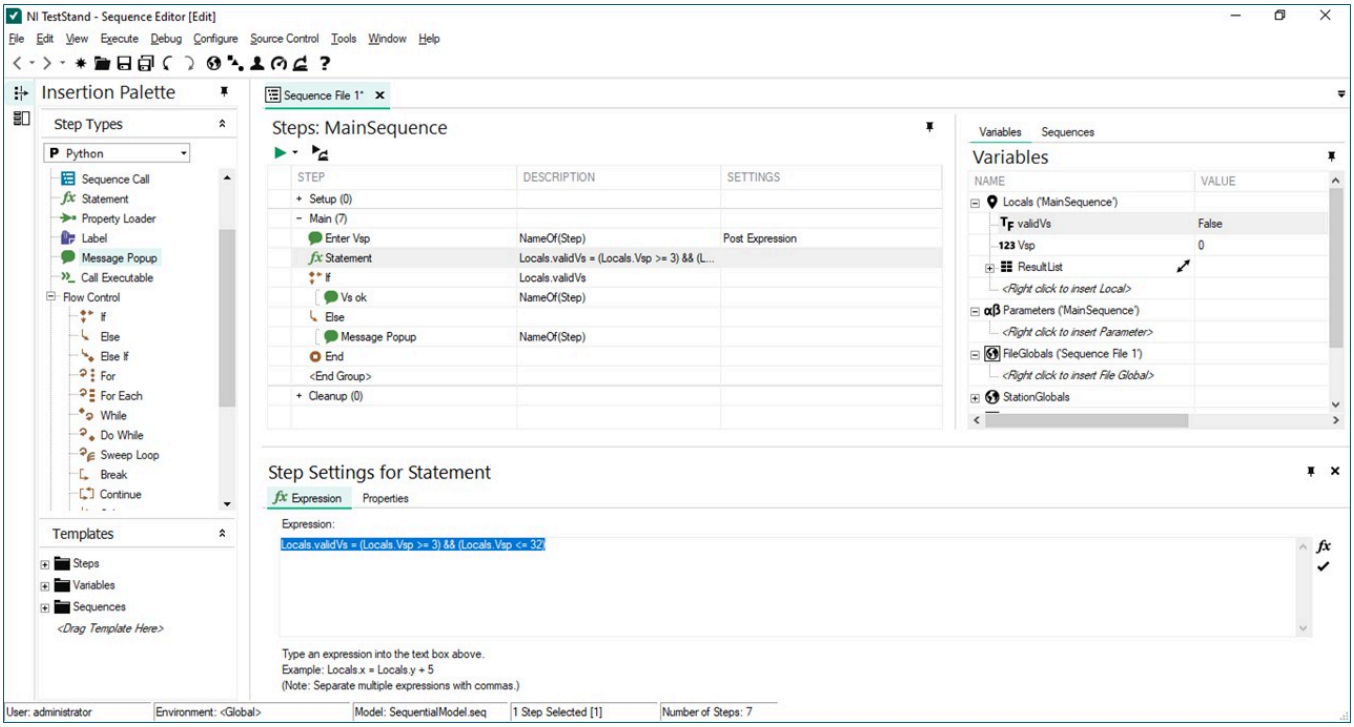


Figure 2. Statement step and If-Else flow control.

2. Validate the value

2.1 Add the Boolean variable **Locals.validVs**.

2.2 Add a **Statement** step: `Locals.validVs = (Locals.Vsp >= 3) && (Locals.Vsp <= 32)`.

2.3 Add **If**, **Else** and **End** flow-control steps. Use `Locals.validVs` as the If condition.

2.4 In the True branch, show “Vs OK”. In the False branch, show “Vs not in range”.

Input	Expected `Locals.validVs`	Message shown	Observed
15	True	Vs OK	
2	False	Vs not in range	
40	False	Vs not in range	
abc			

Question: What does `Val()` return for “abc”, and how should the sequence handle it?

Tutorial 2: Power the AUX op-amp from TestStand

Goal: Call LabVIEW code modules from each step group and pass a VISA reference between them.

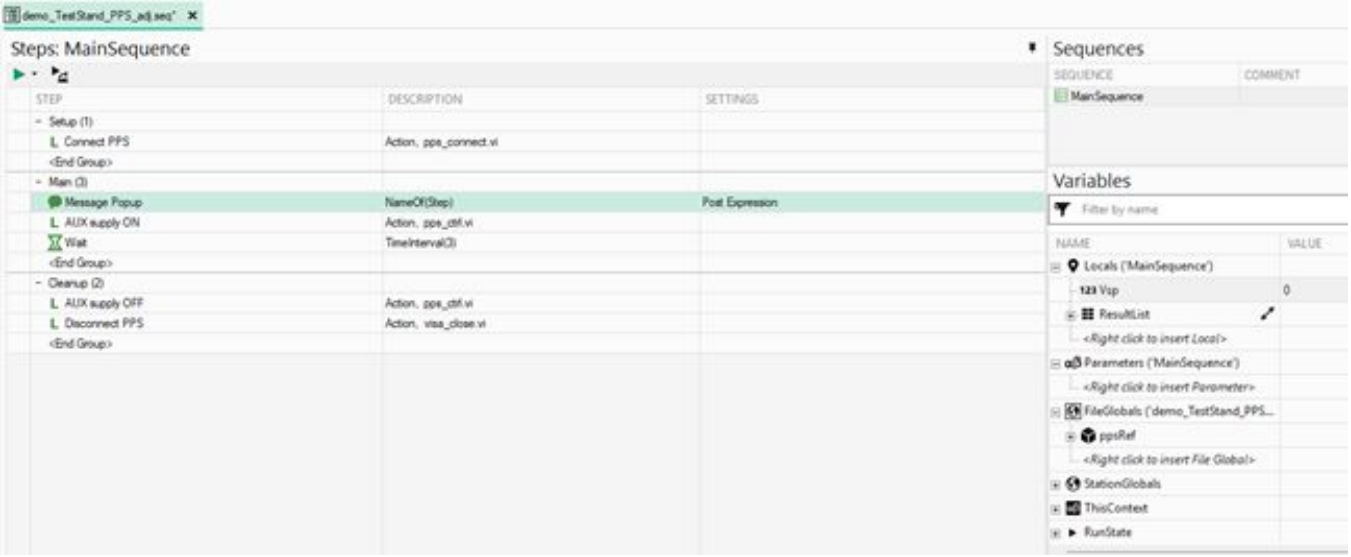


Figure 3. Setup connects the supply, Main applies the voltage for 3 s, Cleanup closes the session.

1. Prepare variables

1.1 Save Tutorial 1 as **Day3_Tutorial2.seq**. Keep the popup that sets **Locals.Vsp**.

1.2 Add **FileGlobals.ppsRef**. Use the LabVIEWIOControl type so it can hold a VISA reference.

2. Add the code modules

2.1 **Setup:** add an Action step that calls **pps_connect**. Map the VISA reference output to `FileGlobals.ppsRef`.

2.2 **Main:** add an Action step that calls **pps_ctrl**. Map the reference input to `FileGlobals.ppsRef` and the voltage input to `Locals.Vsp`.

2.3 **Main:** add a **Wait** step (Synchronization) of 3 s, then call **pps_ctrl** again with 0 V (output off).

2.4 **Cleanup:** call **visa_close** with `FileGlobals.ppsRef`.

3. Run and observe

3.1 With the fixture disconnected or in the approved state, run MainSequence and enter 15.

3.2 Watch the UDP3305S display: output on for about 3 s, then off.

3.3 Run again and press **Terminate** during the Wait. Check that Cleanup still closes the session and the output is off.

Check	Expected	Observed
Display during Wait	15 V, output ON	
Display after Main	output OFF	
After Terminate	session closed, output OFF	

Note: Cleanup closing a session does not by itself switch the output off. If the output stays on after Terminate, add an explicit output-off call to Cleanup before **visa_close**.

Tutorial 3: Supply current I_S test

Goal: Build the full I_S test: four instruments in Setup, a Numeric Limit Test in Main, safe shutdown in Cleanup.

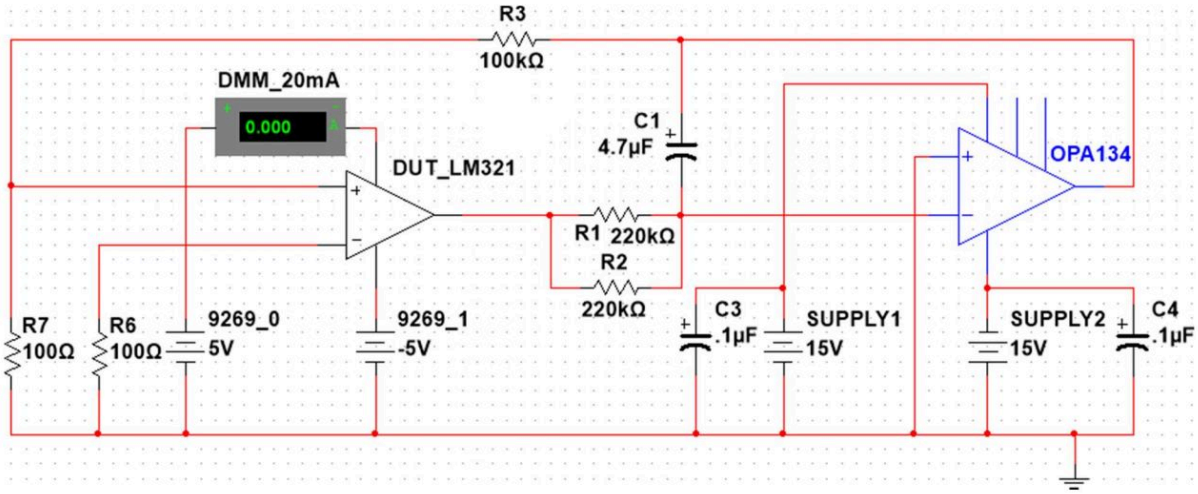


Figure 4. I_S measurement: UDP3305S supplies the AUX op-amp, NI-9269 supplies the DUT, NI-9477 controls S7, UT8805E measures current (20 mA range).

1. Setup group

Code module	Function	Output variable
pps_connect	Connect UDP3305S	`FileGlobals.ppsRef`
dmm_connect	Connect UT8805E	`FileGlobals.dmmRef`
switch_init	Connect NI-9477	`FileGlobals.switchRef`
ao_init	Connect NI-9269	`FileGlobals.aoRef`

1.1 Save as **Day3_LM321_DC.seq**. Add the four FileGlobals (LabVIEWIOControl type).

1.2 Add one Action step per row. Map each reference output to its FileGlobal.

2. Main group

2.1 **pps_ctrl**: AUX supply ±15 V (CH1 = 15 V, CH2 = 15 V as wired on the fixture).

2.2 **switch_ctrl**: S7 ON.

2.3 **ao_update**: DUT supply [5, 5] V on AO0/AO1.

2.4 Add a **Numeric Limit Test** that calls **dmm_read**. Map the current reading to `Step.Result.Numeric`.

2.5 Set the comparison to **LT** with low/high limit 1.2 mA. Use consistent units: if **dmm_read** returns amperes, enter 0.0012.

3. Cleanup group

3.1 **ao_update** [0, 0] V, then **pps_ctrl** output off, then **switch_ctrl** S7 OFF.

3.2 **daqmx_stop** for the NI-9477 and NI-9269 tasks; **visa_close** for the UDP3305S and UT8805E sessions.

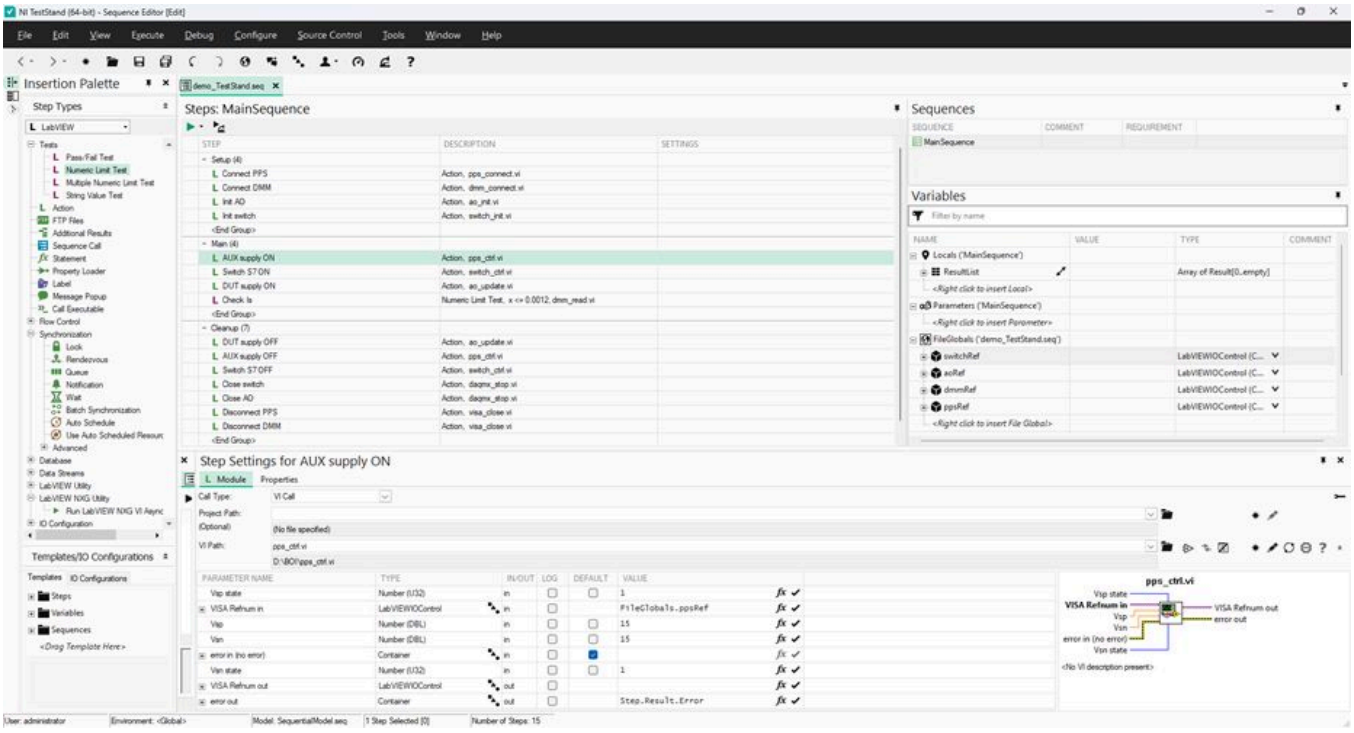


Figure 5. Reference sequence for the I_S test.

4. Run and record

4.1 Run **Execute > Single Pass** so a report is generated. Open the report and find the I_S step.

Item	Value	Unit
Measured I_S		
Limit	< 1.2	mA
Step status (Passed/Failed)		
Datasheet typical I_S (from Figure 6)		

4.2 Lower the limit temporarily below the measured value and run again. Check that the step and the UUT fail, then restore 1.2 mA.

6.1 Absolute Maximum Ratings (1)

	MIN	MAX	UNIT
Differential Input Voltage	±Supply Voltage		
Input Current (V_{IN} < −0.3 V) (2)		50	mA
Supply Voltage (V^+ - V^-)		32	V
Input Voltage	−0.3	32	V

6.5 Electrical Characteristics

Unless otherwise specified, all limits specified for at T_A = 25°C; V^+ = 5 V, V^- = 0 V, V_O = 1.4 V

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{OS}	Input Offset Voltage			(1)2	7	mV
		(1), –40°C ≤ T_J ≤ 85°C			9	
I_{OS}	Input Offset Current			5	50	nA
		–40°C ≤ T_J ≤ 85°C			150	
I_S	Supply Current, No Load	V^+ = 5 V		0.430	1.15	mA
		V^+ = 5 V, –40°C ≤ T_J ≤ 85°C		0.7	1.2	
		V^+ = 30 V		0.660	2.85	
		V^+ = 30 V, –40°C ≤ T_J ≤ 85°C		1.5	3	

Figure 6. LM321 specification.

Tutorial 4: Split the test into sub-sequences

Goal: Move each measurement into its own sequence so MainSequence reads as a test plan, and add the V_{OS} test.

1. Create the I_S sub-sequence

1.1 In **Day3_LM321_DC.seq**, create a new sequence **I_S_test**.

Group	Code module	Function
Setup	**switch_ctrl**	Turn S7 ON
Main	Numeric Limit Test (**dmm_read**)	Test I_S < 1.2 mA
Cleanup	**switch_ctrl**	Turn S7 OFF

1.2 Move the existing steps. The sub-sequence uses the same FileGlobals, so no parameters are needed.

1.3 In MainSequence, replace the moved steps with a **Sequence Call** to **I_S_test**.

2. Create the V_{OS} sub-sequence

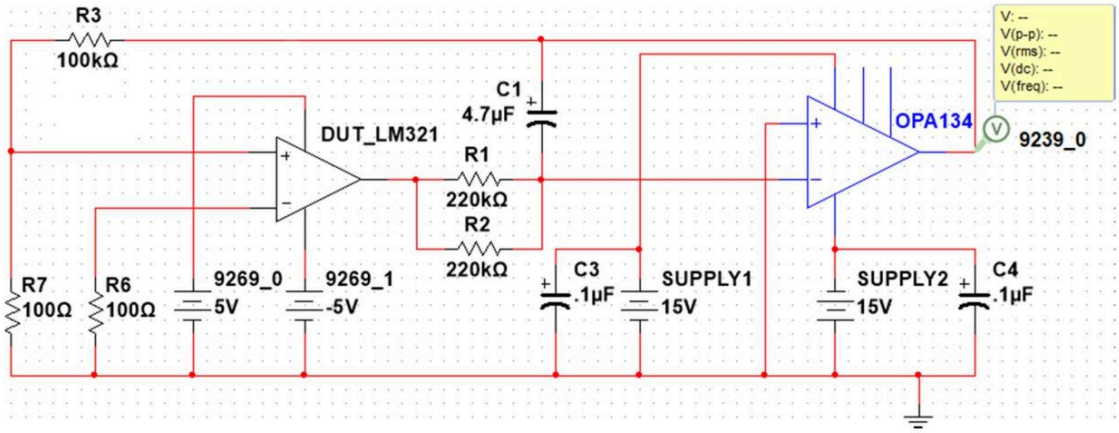


Figure 7. V_{OS} null measurement: the NI-9239 reads the scaled offset at TP1.

2.1 Add **ai_init** to MainSequence Setup → `FileGlobals.aiRef`, and its **daqmx_stop** to Cleanup.

2.2 Create a sequence **V_OS_test** with S7 OFF. In Main, add a **Numeric Limit Test** that calls **ai_measure**.

2.3 Convert TP1 volts to V_{OS} with the station-verified fixture gain (about 1000:1) in a Post-Expression, then test $|V_{OS}| < 7$ mV.

2.4 Add a Sequence Call to **V_OS_test** after **I_S_test**.

3. Final MainSequence

Group	Steps
Setup	Connect UDP3305S, UT8805E, NI-9477, NI-9269, NI-9239
Main	AUX and DUT supplies ON → Call **I_S_test** → Call **V_OS_test**
Cleanup	DUT and AUX supplies OFF → S7 OFF → release DAQmx/VISA resources

4. Record

Test	Measured	Limit	Status
I_S		< 1.2 mA	
V_{OS}		< 7 mV	

Questions: Why does S7 OFF appear in both I_S_test Cleanup and MainSequence Cleanup? What changes if you run with **Test UUTs** instead of Single Pass?

Deliverables

- Day3_Tutorial1.seq, Day3_Tutorial2.seq and Day3_LM321_DC.seq with the VIs they call.
- A Single Pass report showing the I_S and V_{OS} results.

References

Original slides: TS_01_TestStand_overview.md, TS_02_TestStand_workflow.md, TS_03_TestStand_practice.md. Figures are unchanged.

- TestStand Code Module Development Best Practices
- Passing VISA/DAQmx references with LabVIEWIOControl